Technical Report: House Price Prediction System Optimization

Prepared by: N.Dharani

Project: Artificial Intelligence & Machine Learning – Task 2

Dataset: California Housing Dataset

1. Executive Summary

The objective of this project was to expand upon basic machine learning implementations by introducing an industry-aligned machine learning workflow focused on feature scaling, algorithmic comparison, and statistical evaluation. Using the California Housing Dataset, an enhanced House Price Prediction system was developed. Rather than relying on a single algorithm, three distinct machine learning models—Linear Regression, Ridge Regression, and a Decision Tree Regressor—were implemented, optimized, and benchmarked against one another. The final model selection was justified using measurable validation metrics.

2. Methodology & Pipeline Implementation

A structured, reproducible data science pipeline was established to ensure learning stability and prediction accuracy across all models:

- **Data Loading & Inspection:** The California Housing Dataset was imported natively using scikit-learn. The input domain consists of housing-related attributes (such as *Median Income*, *House Age*, and *Average Rooms*) mapped to the target variable, *Median House Value* (HousePrice).

- **Feature Engineering (Scaling):** Numeric ranges varied drastically across features. To prevent features with larger magnitudes from unfairly dominating the loss functions, StandardScaler was applied. This shifted the distributions to a mean of 0 and a variance of 1, facilitating uniform numeric learning.

- **Data Partitioning:** The processed dataset was partitioned using an 80/20 train-test split. This approach ensures that performance evaluations are executed against unseen data, mitigating the risks of model memorization.

- **Algorithmic Diversity:** Three regression models were selected for testing:

1. *Linear Regression:* Serves as the structural baseline.
2. *Ridge Regression ():* Introduces L2 regularization to control weight magnitudes and prevent overfitting.
3. *Decision Tree Regressor ():* Captures non-linear dynamics and split thresholds within the housing features.

3. Results & Performance Comparison

The models were evaluated using two primary industry-standard metrics: **Root Mean Squared Error (RMSE)**, which assesses absolute prediction error, and the **Score**, which evaluates the model's total explanatory power.

The compiled results from the experimental runs are structured in the performance comparison table below:

Model Algorithm	Root Mean Squared Error (RMSE)	Score (Explanatory Power)
Linear Regression (Baseline)	0.555892	0.575788
Ridge Regression	0.555851	0.575819
Decision Tree Regressor	**0.524515**	**0.599732**

Performance Insights:

- **RMSE Analysis:** A lower RMSE points directly to tighter prediction accuracy. The **Decision Tree Regressor** achieved the lowest error score (), outperforming both linear models.
- **Analysis:** A higher score represents a more robust fit. The **Decision Tree Regressor** successfully explained the highest percentage of data variance, yielding an score of (approximately).

4. Final Model Justification & Conclusion

Based on the objective engineering metrics recorded during validation, the **Decision Tree Regressor** was selected as the final deployed model for this system.

Technical Justification: The Decision Tree Regressor outperformed the alternative architectures by securing the lowest overall Root Mean Squared Error of **0.524515** and the highest corresponding score of **0.599732**. While the baseline linear models provided a strong, stable foundation, the Decision Tree's ability to map non-linear structures allowed it to capture complex relationships within the California housing features more effectively. Visual validation via actual vs. predicted scatter plotting confirmed that the model's predictions cluster tightly along the ideal identity reference line, proving its operational readiness and generalization stability on unseen test records.